

```

Terminal
kumargaurav@kumargaurav-VirtualBox: /usr/local/src
Installing DDoS-Deflate 0.6
Downloading source files.....done
Creating cron to run script every minute.....(Default setting).
....done
Installation has completed.
Config file is at /usr/local/ddos.conf
Please send in your comments and/or suggestions to zaf@vsnl.com
kumargaurav@kumargaurav-VirtualBox: /usr/local/src$

```

Figure 3: DDoS Deflate working environment

```

ddos.conf (/usr/local/ddos) - gedit
ddos.conf x
below.
NO_OF_CONNECTIONS=150

##### APF_BAN=1 (Make sure your APF version is atleast 0.96)
##### APF_BAN=0 (Uses iptables for banning ips instead of APF)
APF_BAN=1

##### KILL=0 (Bad Ips are'nt banned, good for interactive
##### KILL=1 (Recommended setting)
KILL=1

##### An email is sent to the following address when an IP is
banned.
##### Blank would suppress sending of mails
EMAIL_TO="root"

##### Number of seconds the banned ip should remain in
blacklist.
BAN_PERIOD=600

```

Figure 4: Configuration file of DDoS Deflate

To view the current status of the daemon and PID running, give the following command:

```
<workingdirectory>$ ddos -t | -status
```

You can view the active connections with the server by typing the following command:

```
<workingdirectory>$ ddos -v | -view
```

The following command bans or blacklists all IP addresses with more than 'n' connections:

```
<workingdirectory>$ ddos -k | -kill:
```

## Fail2Ban

Fail2Ban is another free and open source tool to identify and ban the sources of malicious DDoS traffic. It scans the log files and identifies suspicious patterns and connections so that blacklisting can be done. Fail2Ban reduces the non-legitimate and incorrect authentication attempts with the use of powerful modules for filtering the various services.

The features of Fail2Ban include:

- Deep parsing and analysis of log files
- Awareness of the time zone associated with the source traffic IP

- Integration of client-server architecture
- Assorted services including sshd, vsftpd and Apache can be processed
- Easy configuration for the administrator
- Compatibility with all the firewalls
- Whitelisting and banning of IP addresses
- Blocking of brute force assaults
- Blocking of IP addresses based on time slots
- Excellent for SSH based environments
- Time-based IP blocking
- Support for Python programming

To install and work with Fail2Ban, type the following command:

```
<workingdirectory>$ sudo apt-get install fail2ban
```

Fail2Ban service maintains a configuration file in the directory `/etc/fail2ban`. In this directory, the default configuration file is `jail.conf`.

After installation, the default configuration file is copied to the working configuration file.

```
<workingdirectory>$ sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
```

The configuration settings at the end of the `config` file are as follows:

```

[http-get-dos] # Rule to be Set
enabled = true # Status
port = http,https # 80,443 (Ports)
filter = http-get-dos # Filter Names
logpath = /var/log/www/vhost.d/mysite.com/site-access_log #
Path of Log
maxretry = 5 # Retries Max. Limit
findtime = 10 # 5 retries in 10 seconds from 1 IP Ban or
Blacklist
bantime = 86400 # In Seconds (One Day)
action = iptables[name=HTTP, port=http, protocol=tcp]
        iptables[name=HTTPS, port=https, protocol=tcp]
        sendmail-whois-withline[name=http-get-dos,
dest=<E-mail ID>, logpath=/var/log/httpd/site-access_log] #
sets iptables variables.

```

To view all the `jail` files that have been enabled, type:

```
<workingdirectory>$ sudo fail2ban-client status 
```

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